Video acquisition process — *S. pasteurii*

Problematic

Is it possible to determine with short microscopic video clip the growth phase of S.Pasteurii bacteria, and more specificaly the end of the exponential phase using SSL agorithm ?

Courte vidéo : snapshot de 10 seconde a grossissement x40, microscope photonique
Echantillon : extrait de milieu liquide sur une lamelle.
Milieu liquide : ?
Objectif long terme : mise en place d'un model SSL pour la classification de phase de la bactérie S. Pasteurii, capable de classifier peu importe le milieu.

Objective

Short term: Filming 10 seconds clips of the life cycle of S.Pasteuri from its lag to stationary phase. These clips will be used to develop a SSL model using time-lapse and mobility trackeur (optical flow, tracking).

Long term: The endpoint of our project would be to have a model capable of classifying the bacteria phase in every medium (rich and minimal).

Culture requirements

We must keep in mind that *S. Pasteurii* culture in plain water will not include the exponential phase, lag phase to decline. *S. Pasteurii* needs urea or ammonium and doesn't synthesize by itself essentials nutrients: methionine, cysteine, thiamine and nicotinic acid. In water alone, the *S.Pasteurii* will only show survival to decline. But it can be used as a controlled medium to observe mobility and decline. (Proof)

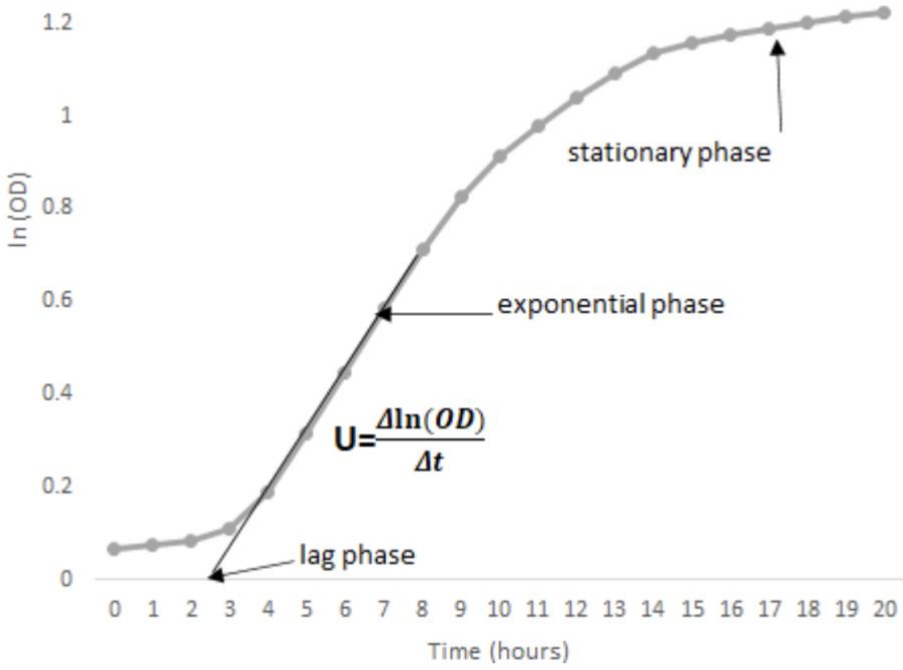
We need to define the different liquid mediums for the life cycle observations. I listed below existing mediums in literature as a base with the corresponding duplication time observed to further determine the most appropriates solutions.

	YE		NH4-YE			DSMZ Nut		
	No CO_2	10vol% CO_2	No CO_2	1vol% CO_2	10vol% CO_2	No CO_2	1vol% CO_2	10vol% CO_2
OD_0 (t=0)	0.08	0.07	0.09	0.09	0.06	0.06	0.07	0.06
OD_{max}	0.56	0.68	1.41	1.22	0.09	1.05	1.65	0.30
OD_{max}-OD_0	0.48	0.61	1.31	1.13	0.02	0.99	1.58	0.23
U (h^{-1})	0.25	0.24	0.20	0.27	0.14	0.29	0.38	0.03
Td (h)	2.73	2.90	3.50	2.57	4.88	2.44	1.84	23.42

	DSMZ 5g/L			DSMZ 20g/L			DSMZ 50g/L		
	No CO_2	1vol% CO_2	10vol % CO_2	No CO_2	1vol% CO_2	10vol % CO_2	No CO_2	1vol% CO_2	10vol % CO_2
OD_0 (t=0)	0.06	0.07	0.06	0.06	0.07	0.07	0.07	0.06	0.06
OD_{max}	1.40	1.57	1.09	1.40	1.54	1.40	1.28	1.22	0.86
OD_{max}-OD_0	1.34	1.50	1.03	1.34	1.48	1.33	1.21	1.16	0.79
U (h^{-1})	0.38	0.45	0.20	0.61	0.54	0.43	0.50	0.44	0.40
Td (h)	1.81	1.54	3.40	1.14	1.29	1.62	1.40	1.59	1.73

Every measurement has been done with 96 plaques cultures at 30° and 200 rpm. (Article Download link)

Optimal Video Clips Calendar



Typical bacterial growth curve: ln OD is plotted versus time. (download link, page7)

From this typical bacteria growth curve, we can build some sort of data acquisition patterns based of nyquist sampling logic. But since growth curve isn't a periodic signal it will not be the perfect data collection technique.

Exponential phase typicaly (in rich medium)last for 9 to 12 hours, before oxygen deprivation cause the stationary phase. This is the phase where we aim to have the most records/clips to identify the determinant switch use by the MICP.

Process discussion

I don't think that I will be able to provide the perfect plan to grow the bacteria in different medium and gather the data, but I needed to list every needs that we have to ensure the best output.

We need to decide the number of different cultures and the rate of recordings for each phase.
Here is an example of data gathering cylces depending on the medium and phase

Medium	Lag + expo	stationary	Death	End
DSMZ 220 + urea	Each 30 min, for 0-14 h	Each 2 h, for 14-24 h	Each 12 h	96 h
DSMZ 220 wihtout urea	Each 1 h, for 0-20 h	Each 3 h, for 20-36 h	Each 12 h	96 h
NH4-YE	Each 2 h, for 0-30 h	Each 4 h, for 30-48 h	Each 12 h	120 h
Plain water	Each 12 h	Each 12 h	Each 12 h	7 days

S. Pasteurii Mobility Discussion

We didn't find any recording/measurements of the bacteria volicity. We only know it depends on the medium they are growing in and is related to their global energy level.

State	**Approximate Mobility**
Exponential, medium with urea	Max at 15–25 µm/s for *B. subtilis*
End of exponential / Stationary	Decreasing, less mobile cells, chains of bacteria are forming
Death/water	Mostly Brownian

Sizing: number of clips per phase

Follow 12 cultures per medium from inoculation to decline, recording 3 fields × 10 s at each time point of the schedule. For DSMZ 220 + urea, this gives:

Phase	**Videos per culture**	**Approximate Mobility**
Lag	~15	~180
Exponential	~72	~860
Stationary	~18	~215
Decline	~18	~215

Do not add fields to balance the phases: if data is missing, add cultures.

Sources

- Ennaciri, Clarà Saracho & Marek 2022 — temps de doublement par milieu
- Lapierre et al. 2020, Sci Rep — besoins nutritionnels et cinétique de DSM 33
- Ma et al. 2020, Microb Cell Fact — effet de l'urée sur la croissance
- Thèse KU — effet de la température et de l'urée